

Physical Chemistry

1. Introduction

Physical chemistry explains the laws and principles that govern the behavior of matter, energy, and chemical change. It combines thermodynamics, quantum theory, kinetics, electrochemistry, phase equilibria, spectroscopy, surface chemistry, and statistical ideas to explain why reactions occur and how fast they proceed.

At the BSc level, physical chemistry is one of the most mathematically and conceptually important branches of chemistry because it connects microscopic molecular behavior with macroscopic observables such as pressure, temperature, conductivity, and equilibrium.

2. States of matter

The study of states of matter begins with gases, liquids, and solids, and often includes intermolecular forces and transport properties. Gas laws, kinetic molecular theory, real gases, and deviations from ideality are core topics.

Gas laws

- Boyle's law.
- Charles's law.
- Avogadro's law.
- Combined gas law.
- Ideal gas equation $PV = nRT$.

Kinetic molecular theory

- Gases consist of particles in constant motion.
- Collisions are elastic.
- Particle volume is negligible in the ideal model.
- Intermolecular forces are negligible for ideal gases.

Real gases

Real gases deviate from ideal behavior at high pressure and low temperature. The van der Waals equation corrects for molecular size and intermolecular attractions.

3. Thermodynamics

Thermodynamics is the study of energy changes in chemical and physical processes. It provides the foundation for understanding spontaneity, equilibrium, heat, work, entropy, and free energy.

Basic terms

- System.
- Surroundings.
- Boundary.
- State function.
- Path function.
- Intensive and extensive properties.

First law

The first law states that energy is conserved. Internal energy change is related to heat and work.

Enthalpy

Enthalpy is defined as $H = U + PV$. It is especially useful at constant pressure.

Thermochemistry

Thermochemistry deals with heat changes accompanying reactions. Important ideas include:

- Standard enthalpy of formation.
- Enthalpy of combustion.
- Enthalpy of neutralization.
- Enthalpy of solution.
- Hess's law.
- Kirchhoff's equation.

4. Second laws

The second law introduces entropy and the direction of spontaneous change. It explains why some processes occur naturally and others do not. Entropy is a measure of disorder or energy dispersal.

Key points

- Entropy is a state function.
- Spontaneous processes tend to increase total entropy.
- Free energy combines enthalpy and entropy effects.
- Equilibrium occurs when driving forces balance.

5. Chemical equilibrium

Chemical equilibrium is a dynamic state in which forward and reverse reaction rates are equal. It is governed by equilibrium constants and thermodynamic relationships.

6. Phase equilibria

Phase equilibria examines the equilibrium between different phases of matter. It includes phase diagrams, phase rule, and transition points.

Gibbs phase rule

$F = C - P + 2$, where F is degrees of freedom, C is components, and P is phases.

Applications

- Water system.
- Sulphur system.
- Eutectic mixtures.
- Binary phase diagrams.

7. Solutions

Solutions are homogeneous mixtures and are very important in physical chemistry. Their behavior is described by concentration terms, colligative properties, and solution thermodynamics.

Concentration units

- Molarity.
- Molality.
- Mole fraction.
- Normality.
- Mass percentage.

Colligative properties

- Relative lowering of vapor pressure.
- Elevation of boiling point.
- Depression of freezing point.
- Osmotic pressure.

8. Chemical kinetics

Chemical kinetics studies the rate of chemical reactions and the mechanism by which reactions occur. It is one of the most important core units in physical chemistry.

Mechanistic ideas

- Collision theory.
- Transition state theory.
- Steady-state approximation.
- Rate-determining step.
- Kinetic and thermodynamic control.

Catalysis

Catalysis may be homogeneous, heterogeneous, or enzymatic. Catalysts change reaction rate without being consumed.

9. Electrochemistry

Electrochemistry deals with the relationship between electrical energy and chemical change. It includes galvanic cells, electrolytic cells, electrode potentials, conductivity, and electrolysis.

10. Quantum chemistry

Quantum chemistry explains atomic and molecular behavior using wave mechanics. It is necessary for understanding orbitals, energy quantization, and molecular structure.

11. Atomic and molecular structure

This unit explains how atoms are arranged and how molecules form. It includes hydrogen atom theory, quantum numbers, electronic configuration, bonding theories, and molecular orbital ideas.

12. Surface chemistry

Surface chemistry studies adsorption, catalysis, colloids, emulsions, micelles, and interfacial phenomena. It is important in industrial and biological processes.

13. Spectroscopy

Spectroscopy helps identify structure through interaction of matter with electromagnetic radiation. It is a major advanced topic in many BSc syllabi.

Common types

- UV-Visible spectroscopy.
- Infrared spectroscopy.
- NMR spectroscopy.
- Mass spectrometry.
- Rotational and vibrational spectroscopy in advanced courses.

14. Key formulas to remember

- Ideal gas law: $PV = nRT$.
- Gibbs phase rule: $F = C - P + 2$.
- Arrhenius equation: $k = Ae^{-E_a/RT}$.
- Nernst equation for electrochemical cells.
- Bond-related and equilibrium relations in thermodynamics and kinetics.

15. Important comparison points

Thermodynamics vs kinetics

- Thermodynamics tells whether a reaction is feasible.
- Kinetics tells how fast it occurs.

State function vs path function

- State function depends only on initial and final states.
- Path function depends on the route taken.

Physisorption vs chemisorption

- Physisorption is weak and reversible.
- Chemisorption is stronger and often specific.

16. Summary

Physical chemistry is the branch that connects laws of energy, matter, structure, and reaction behavior into one quantitative framework. For Bsc Chemistry, it is the backbone of theory-based chemistry because it explains equilibrium, kinetics, thermodynamics, electrochemistry, and molecular structure in a unified way.